

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

II
CHOMSKY NORMAL FORM (CNF)

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REVIEW

Simplification of CFG:

Elimination of Useless Productions

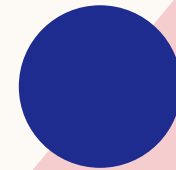
Elimination of Unit Productions

Elimination of Null (ε) Productions

OUTLINE

Chomsky Normal Form (CNF)

Conversion of CFG to CNF



CHOMSKY NORMAL FORM (CNF)

- To put a CFG in CNF, it must satisfy the restrictions:
 - The CFG has no ε -productions
 - The CFG has no unit productions
 - The CFG has no useless productions / symbols
- A simplified CFG in which all productions are in one of two simple forms, either:
 - $A \rightarrow BC$, where A, B , and C , are each variables
 - $A \rightarrow a$, where A is a variable and a is a terminal
- Example
 - $C \rightarrow BA \mid d$

CONVERSION OF CFG TO CNF

- I. Arrange that all bodies of length 2 or more consist only of variables
 - For every terminal a that appears in a body of length 2 or more, create a new variable, A
 - This variable has only one production, $A \rightarrow a$
 - Use A in place of a everywhere a appears in a body of length 2 or more

CONVERSION OF CFG TO CNF

2. Break bodies of length 3 or more into a cascade of productions, each with a body consisting of two variables
 - Break those productions $A \rightarrow B_1 B_2 \dots B_k$, for $k \geq 3$, into a group of productions with two variables in each body
 - Introduce $k - 2$ new variables, $C_1, C_2, \dots, C_{k-2}$
 - The original production is replaced by the $k - 1$ productions
 - $A \rightarrow B_1 C_1$
 - $C_1 \rightarrow B_2 C_2$
 - ...
 - $C_{k-3} \rightarrow B_{k-2} C_{k-2}$
 - $C_{k-2} \rightarrow B_{k-1} B_k$

CONVERSION OF CFG TO CNF

■ CFG

- $S \rightarrow bA \mid aB$
- $A \rightarrow bAA \mid aS \mid \mathbf{a}$
- $B \rightarrow aBB \mid bS \mid \mathbf{b}$

■ CNF

- $S \rightarrow P_1A \mid P_2B$
- $A \rightarrow P_1P_3 \mid P_2S \mid \mathbf{a}$
- $B \rightarrow P_2P_4 \mid P_1S \mid \mathbf{b}$
- $P_1 \rightarrow b$
- $P_2 \rightarrow a$
- $P_3 \rightarrow AA$
- $P_4 \rightarrow BB$

CONVERSION OF CFG TO CNF

■ CFG

- $S \rightarrow aB \mid \mathbf{CA}$
- $A \rightarrow \mathbf{a} \mid bc$
- $B \rightarrow \mathbf{BC} \mid Ab$
- $C \rightarrow aB \mid \mathbf{b}$

■ CNF

- $S \rightarrow P_1B \mid \mathbf{CA}$
- $A \rightarrow \mathbf{a} \mid P_2P_3$
- $B \rightarrow \mathbf{BC} \mid AP_2$
- $C \rightarrow P_1B \mid \mathbf{b}$
- $P_1 \rightarrow a$
- $P_2 \rightarrow b$
- $P_3 \rightarrow c$

CONVERSION OF CFG TO CNF

CFG

- $S \rightarrow aAB \mid ch \mid \mathbf{CD}$
- $A \rightarrow dbE \mid eEC$
- $B \rightarrow ff \mid \mathbf{DD}$
- $C \rightarrow ADB \mid aS$
- $D \rightarrow \mathbf{i}$
- $E \rightarrow jD$

CNF

- $S \rightarrow P_1P_2 \mid P_3P_4 \mid \mathbf{CD}$
- $A \rightarrow P_5P_6 \mid P_8P_9$
- $B \rightarrow P_{10}P_{10} \mid \mathbf{DD}$
- $C \rightarrow AP_{11} \mid P_1S$
- $D \rightarrow \mathbf{i}$
- $E \rightarrow P_{12}D$
- $P_1 \rightarrow a$
- $P_2 \rightarrow AB$
- $P_3 \rightarrow c$
- $P_4 \rightarrow h$
- $P_5 \rightarrow d$
- $P_6 \rightarrow P_7E$
- $P_7 \rightarrow b$
- $P_8 \rightarrow e$
- $P_9 \rightarrow EC$
- $P_{10} \rightarrow f$
- $P_{11} \rightarrow DB$
- $P_{12} \rightarrow j$



PRACTICE

EXERCISES

1. Put the CFG below into CNF.

$$S \rightarrow aSb \mid ab$$

2. Put the CFG below into CNF.

$$S \rightarrow aSaA \mid A$$

$$A \rightarrow abA \mid b$$

EXERCISES

3. Put the CFG below into CNF.

$$S \rightarrow abAB$$

$$A \rightarrow bAB \mid \varepsilon$$

$$B \rightarrow BAa \mid A \mid \varepsilon$$

4. Put the CFG below into CNF.

$$S \rightarrow aB \mid aaB$$

$$A \rightarrow \varepsilon$$

$$B \rightarrow bA \mid \varepsilon$$

EXERCISES

5. Put the CFG below into CNF.

$$S \rightarrow ABaC$$

$$A \rightarrow Bd$$

$$B \rightarrow b \mid \varepsilon$$

$$C \rightarrow D \mid \varepsilon$$

$$D \rightarrow BCa$$

REFERENCES

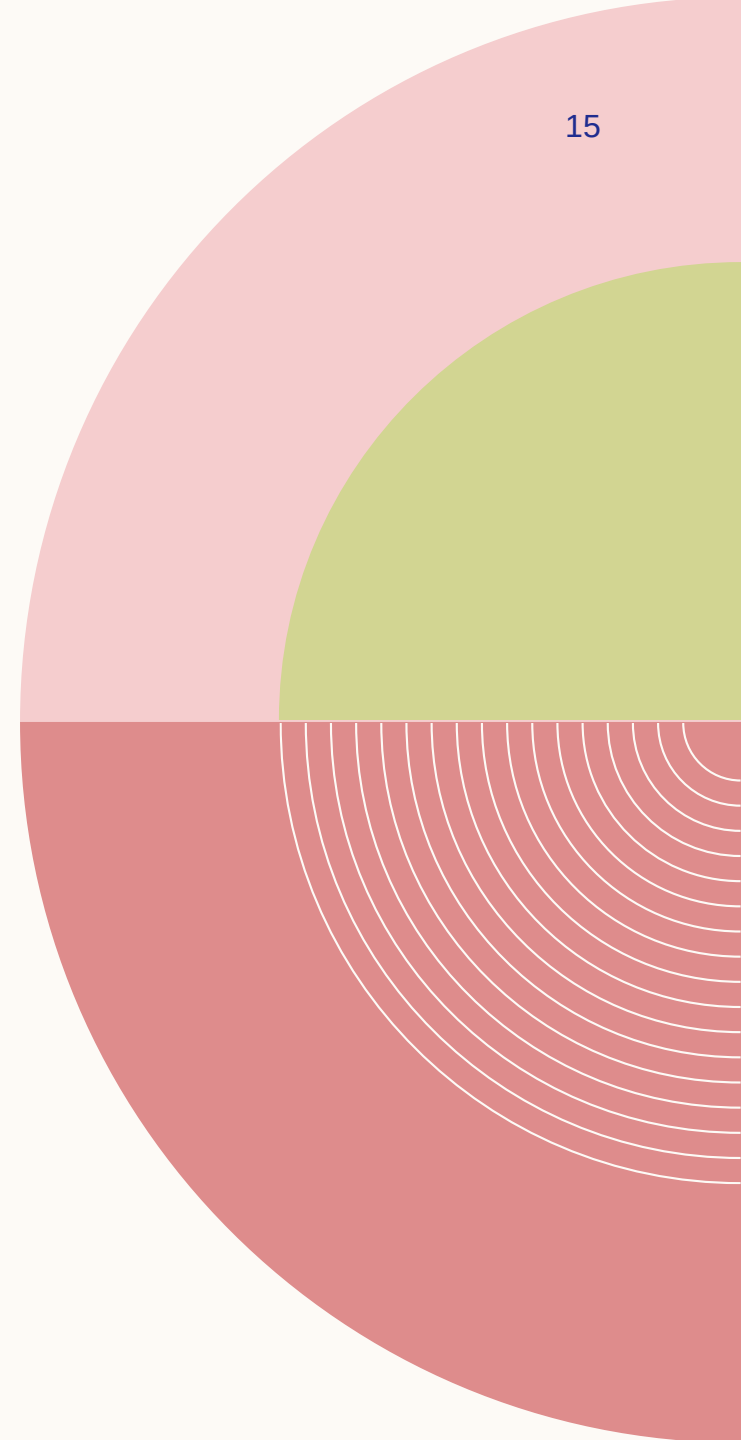
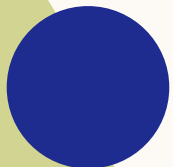
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NEXT

Elimination of Left Recursion:

Recursive Production Rules

Elimination of Left Recursion



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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